

STT 890-003: Introduction To MLOps

Course Syllabus

Fall 2026

Instructor: Paul Speaker

Locations: Engineering Building 3400

Hours: TTh 1:00-2:20

Course D2L page: <https://d2l.msu.edu/d2l/home/2509832>

Office Hours: TBD

Textbook: None

Course Objectives: The purpose of this course is to give students thorough introduction to Machine Learning Operations (MLOps) in practice. After completing this course, students will:

- (1) Understand the full process of working with Data Science models, from stakeholder assessment, data intake and analysis, productionizing models, to model monitoring and lifecycle.
- (2) Be able to plan Data Science projects and understand and anticipate what problems can occur throughout the lifecycle process.
- (3) Understand the data processing needed to get to modeling work.
- (4) Understand the constraints that the productionizing of models places on modeling work.
- (5) Be able to use technical tools for working within MLOps
- (6) Be able to use Docker for productionizing models
- (7) Be able to use technical tests for assessing model performance once the model is being used.

Course Grades:

(1) Homework:	20%
(2) Exams:	25%
(3) In-Class activities:	25%
(4) Project:	30%

1. **Homework.** There will be 4 homework assignments to be turned in. Appropriate libraries and commands that are specific to probability and statistics, as well as some plotting, will be pointed out during the lectures. You will turn in both the code and the output in Jupyter notebooks, with a pdf export. The lowest grade of the assignments will be dropped. Late homework will not be accepted.

2. **Exams.** There will be 2 exams during the semester. The format of these exams will vary. Any written exam will be in-class, while exams in other format will be scheduled by sign-up.
3. **In-Class Assignments.** Almost all classes will have in-class activities. Work on these done outside of class will not be accepted. Three such grades will be dropped.
4. **Project.** The project will involve the creation of an end-to-end ML solutions through MLOps. Various steps will be further detailed in the class. Everyone will be involved in two projects. First, everyone (in teams of 2) will decide on a project as a developer. Second, everyone will be assigned to someone else's project as a user, where they will evaluate and stress test the project. More details will be given later.

Grade Assignment (Grading Scale):

Grade	Percentage
4.0	90-100%
3.5	85-89.9%
3.0	80-84.9%
2.5	75-79.9%
2.0	70-74.9%
1.5	65-69.9%
1.0	60-64.9%
0.0	< 60%

Tentative Course Schedule

Week of	Tuesday	Thursday
8/31/26	Course Intro	The Ask
9/7/26	Who Does What	Inertia and How to Overcome
9/14/26	Success and its Measurement	Impossible Asks
9/21/26	Planning ML Projects	No Class
9/28/26	Process Risk	Exam I
10/5/26	Intro to DevOps	Static Data and Other Lies
10/12/26	Explicit and Implicit Target Leakage	Data Pipelines
10/19/26	ML Reproducibility	From Dev to Prod
10/26/26	No Class-Fall Break	Testing Productionization
11/2/26	Exam II	Docker
11/9/26	Kubernetes	Dashboards
11/16/26	Work Process Redesign	Model Maintenance and Data Drift
11/23/26	Changes in Model Performance	No Class--Thanksgiving
11/30/26	Model Monitoring Tools	Model Monitoring and UX
12/7/26	Project Presentations	Project Presentations

Required Technologies:

- Computer, with Anaconda installed. We will be using Python for the class.

COVID Protocols

“Students are required to follow MSU protocols after a positive test for COVID or experiencing COVID symptoms. The protocols can be found here: <https://msu.edu/together-we-will/covid19-guidance/>

Learning Continuity Statement:

In the event you are unable to attend class for an extended period of time, please let me know as soon as possible. Alternative accommodations can be given in this event for homework and other assessments.

Student Expectations

The All-University Policy on Integrity Of Scholarship and Grades:

All participants in this class are held to the standard set by MSU’s Policy on Integrity of Scholarship and Grades. The policy can be read in full at the [MSU Ombudsperson’s website](#)

Spartan Code of Honor:

On March 22, 2016, The Associated Students of Michigan State University (ASMSU) adopted the following Spartan Code of Honor:

“As a Spartan, I will strive to uphold values of the highest ethical standard. I will practice honesty in my work, foster honesty in my peers, and take pride in knowing that honor is worth more than grades. I will carry these values beyond my time as a student at Michigan State University, continuing the endeavor to build personal integrity in all that I do.”

Disability Access:

Students must inform the instructor of any accommodations needed. Information related to disability access is available on the [Resource Center for Persons with Disabilities \(RCPD\) website](#). Students: to make an appointment with a specialist, call: (517) 353-9642 Or TTY: (517) 355-1293 or visit the [RCPD website](#).

Americans with Disabilities Act Accommodations:

- Include a statement inviting ADA accommodation requests. [Statement on ADA Compliance](#): *This is your opportunity to describe what efforts you have made to ensure compliance with the Americans with Disabilities Act for the online version or portion of*

the course and how people with disabilities can contact you and RCPD if aspects of the course are not accessible due to a disability.

Sample accessibility statement: [RCPD Disability Accommodations Statement](#)

“Michigan State University is committed to providing equal opportunity for participation in all programs, services and activities. Requests for accommodations by persons with disabilities may be made by contacting the Resource Center for Persons with Disabilities at 517-884-RCPD or on the web at rcpd.msu.edu. Once your eligibility for an accommodation has been determined, you will be issued a verified individual services accommodation (“VISA”) form. Please present this form to me at the start of the term and/or two weeks prior to the accommodation date (test, project, etc). Requests received after this date will be honored whenever possible.”

Grief Absences and Mental Health:

If a student experiences death of a family member or emotional distress from a similar tragedy, refer to [MSU's Grief Absense Policy](#) (<https://msu.edu/unit/ombud/classroomolicies/index.html#GriefAbsencePolicy>).

· For Mental Health resources see: [Mental Health](#)

Commercialized Lecture Notes: Commercialization of lecture notes and university-provided course materials is *not permitted* in this course.

Student Rights and Responsibilities:

Students have a range of support and information options available to them to discuss actions or activities related to their academic, personal or professional lives at MSU. These rights and resources are detailed on the [ombudsperson's website](#).